Repeating Efficiently

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Repeating Efficiently before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

If/Then Decisions

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features: Useful Clues

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pictures Become Numbers

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Repeating Efficiently trace and record the first step you can explain without help.

Use W2: Check the first test for **Loop**s and stopping rules. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Repeating Efficiently explanation against the supplied facts before discussing it.

Use the worked case for **Loop**s and stopping rules. Proposed explanation: “Every **loop** runs at least once”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer to a countdown

A paper countdown starts at 7. While remaining is above 1, subtract 2. When false, report and stop. Trace the run and count **iteration**s. Repeat from 6 and from 1. Explain why the final value need not equal 1.

What happens from 7 if subtract 2 is replaced by subtract 0?

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Repeating Efficiently target to a trace and a justified conclusion.

A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Repeating Efficiently case and show the deciding evidence.

Repeat take the next unstamped card and stamp it four times.

How many cards are stamped? What assumption matters?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Repeat clap then tap three times. Write the complete sequence. How many actions occur? What would happen if the stopping instruction were missing?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Repeating Efficiently adds to Helpful Classroom Sorter and keep the evidence.

Use a **loop** to **process** each item card in the test pile, with an explicit stop when no cards remain.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Trace every test

> Trace every test. Apply the supplied Repeating Efficiently facts and keep your reasoning.

Start counter at 2. While counter is below 8, add 2. When false, report and stop. Record the value and result at every test, including the final false test. State the number of **iteration**s and the reported value.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Check the first test

> Check the first test. Apply the supplied Repeating Efficiently facts and keep your reasoning.

Use the same below 8 **loop** with add 2. Make separate runs from 8 and 9. Explain how many times the **body** runs and what each run reports. Why would adding 2 once be an error?

My reasoning and evidence

W3 Find the extra repetition

> Find the extra repetition. Apply the supplied Repeating Efficiently facts and keep your reasoning.

Start counter at 3. While counter is below 8, add 2. A learner writes values 3, 5, 7, 9, 11 and reports 11. Identify the first unauthorised update. Give the correct final test, reported value and **iteration** count.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer to a countdown

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer to a countdown. Apply the supplied Repeating Efficiently facts and keep your reasoning.

A paper countdown starts at 7. While remaining is above 1, subtract 2. When false, report and stop. Trace the run and count **iteration**s. Repeat from 6 and from 1. Explain why the final value need not equal 1.

What happens from 7 if subtract 2 is replaced by subtract 0?

> **My reasoning and evidence**

WORKBOOK · 16 OF 21

W5 Test a proposed stopping rule

> Test a proposed stopping rule. Apply the supplied Repeating Efficiently facts and keep your reasoning.

A procedure says start at 4; while counter is not equal to 9, add 2; report and stop. Write the first four values tested. Explain why the **loop** will not reach 9 using whole numbers.

Suggest a changed condition that would stop once the counter reaches or passes 9, and trace that repaired run.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Repeating Efficiently

> A fictional paper counter advances through a display until its condition becomes false.

> Explain the labelled step before checking the changed condition.

A fictional paper counter advances through a display until its condition becomes false.

Start at 2. While the current counter is below 10, add 3.

Test before each addition; report when the condition is false.

AI ACADEMY / AI-1/11

Condition checks before additions

	Current	Below 10?	Next
Check 1	2	true	5
Check 2	5	true	8
Check 3	8	true	11
Check 4	11	false	stop

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/11

Repeating Efficiently

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First test

2<10 is true: update to 5.



2. Repeat

5<10: update to 8. 8<10: update to 11.



3. Stop test

11<10 is false; do not add again.



4. Report

Final counter 11 after three additions.

CHANGE ONE THING

Start at 10 instead of 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace a fresh start of 9. How many additions occur, and must the report equal 10?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper counter advances through a display until its condition becomes false.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Final counter 11 after three additions.

1. Report

2. First test

3. Repeat

4. Stop test

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report Final counter 11 after three additions. The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10.

The final 11 is therefore allowed by this procedure. Record both the condition and counter to distinguish a correct stop from an assumed endpoint.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Start at 10 instead of 2. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The **loop** runs zero times and reports 10.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **loop** runs zero times and reports 10. Its first condition is already false, so no addition happens.

Repair a missing building block

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

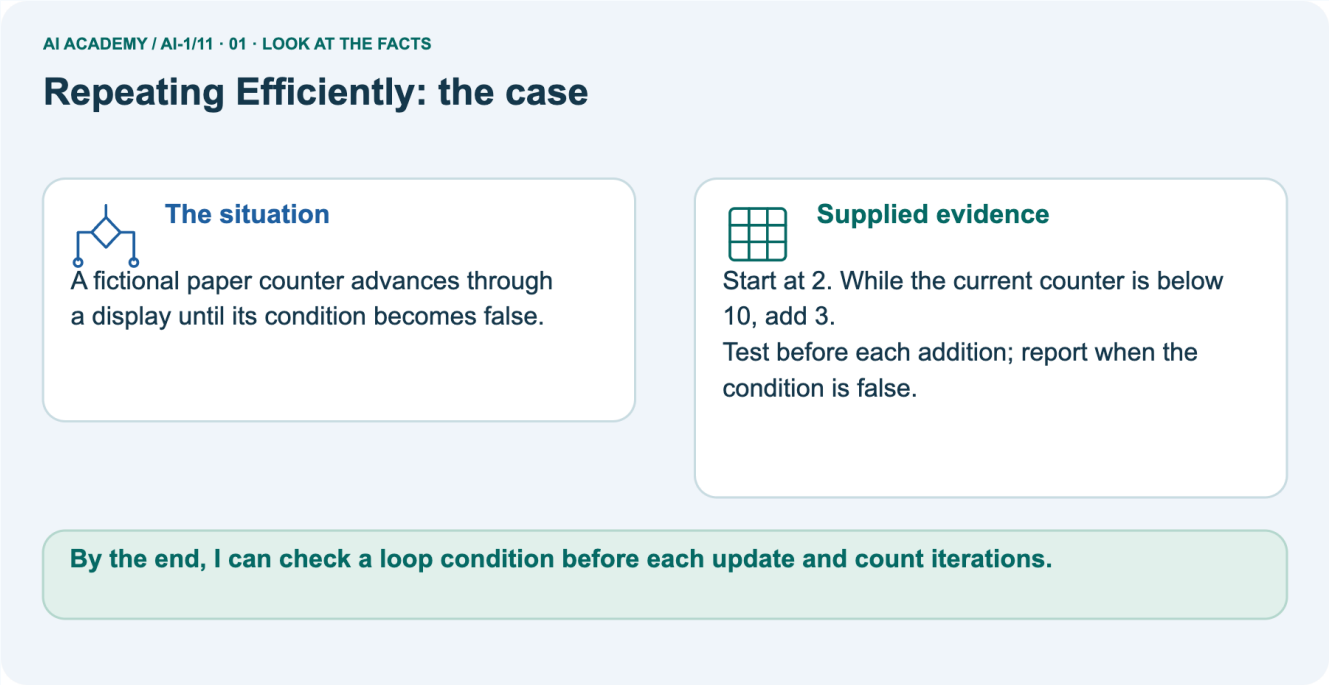
> **What I repaired, the evidence I kept and the next test**
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> | |
> |---|
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WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Repeating Efficiently picture from its labels, then record a relevant limit.

01. Repeating Efficiently: the case



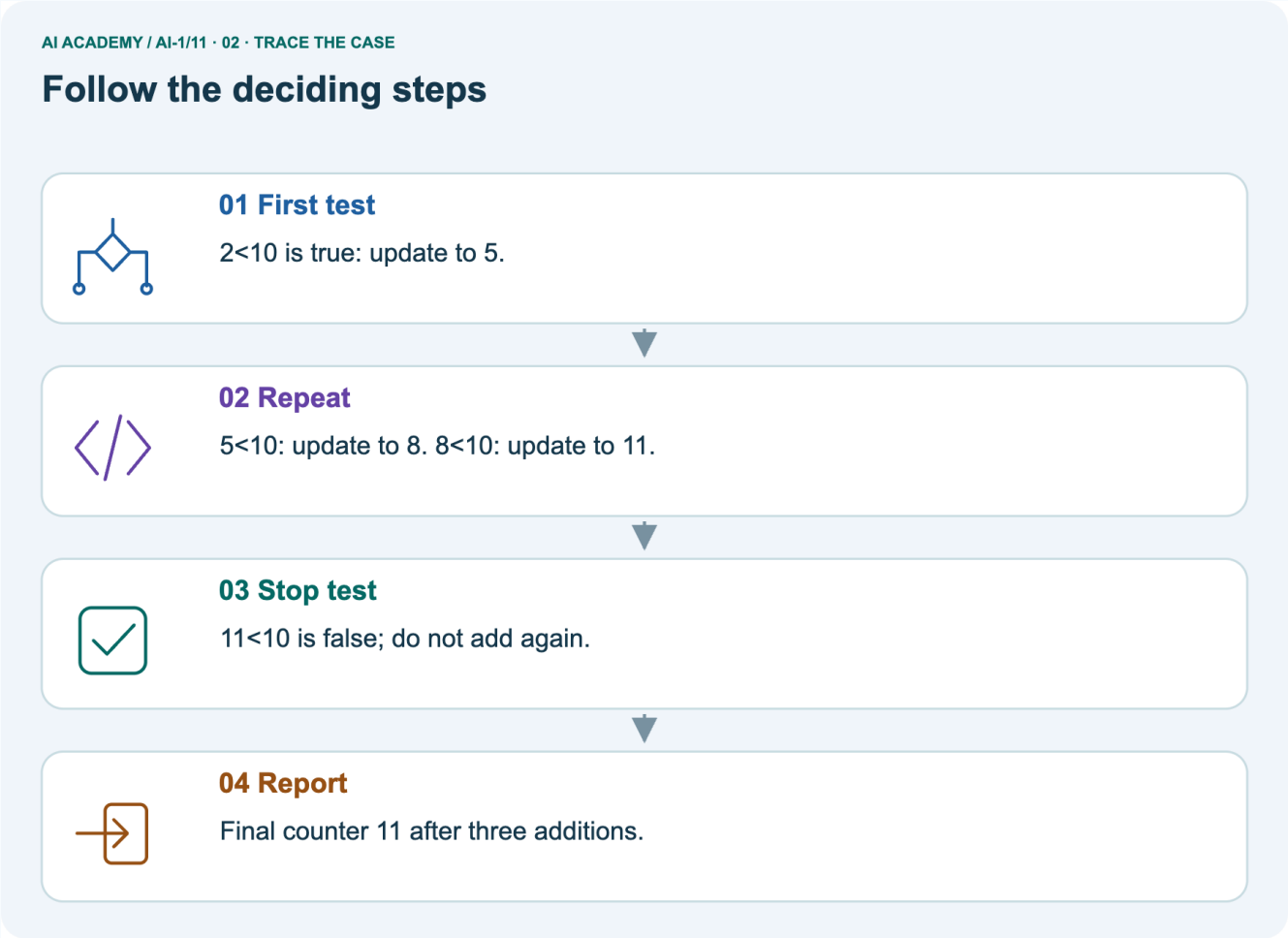
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper counter advances through a display until its condition becomes false. Supplied facts: Start at 2. While the current counter is below 10, add 3. Test before each addition; report when the condition is false.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



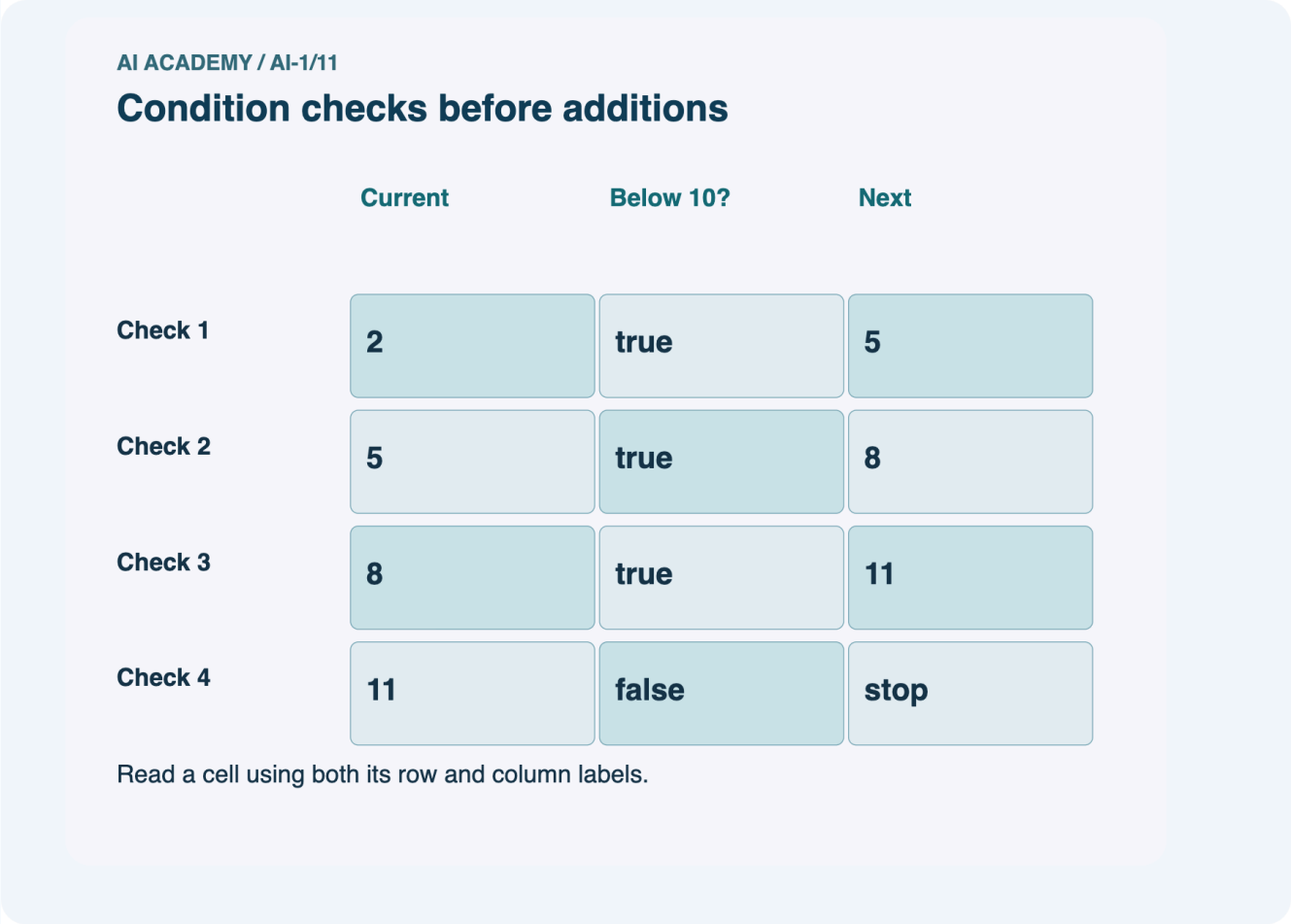
Follow the labelled picture with the matching explanation. Open the vector diagram

First test: 2<10 is true: update to 5. → Repeat: 5<10: update to 8. 8<10: update to 11. → Stop test: 11<10 is false; do not add again. → Report: Final counter 11 after three additions.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Condition checks before additions



Follow the labelled picture with the matching explanation. Open the vector diagram

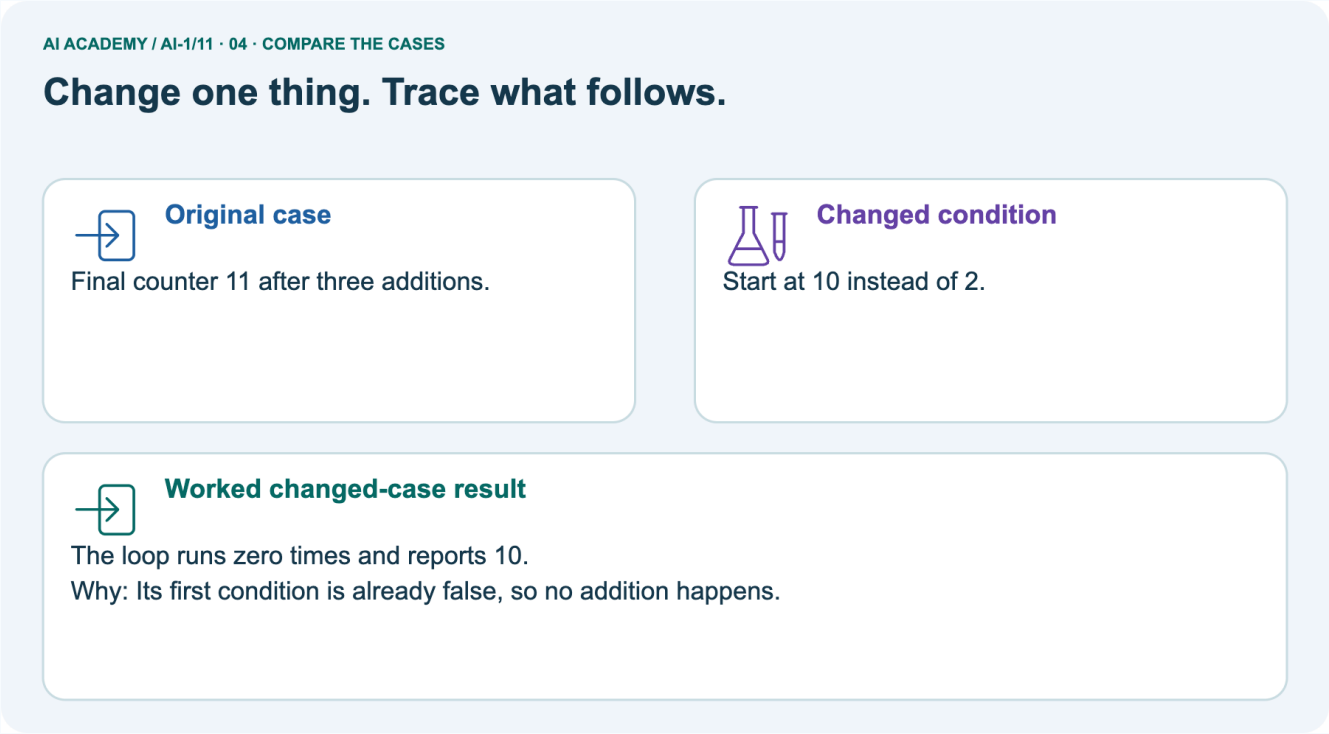
Condition checks before additions. Check 1: Current = 2, Below 10? = true, Next = 5. Check 2: Current = 5, Below 10? = true, Next = 8. Check 3: Current = 8, Below 10? = true, Next = 11.

Check 4: Current = 11, Below 10? = false, Next = stop.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

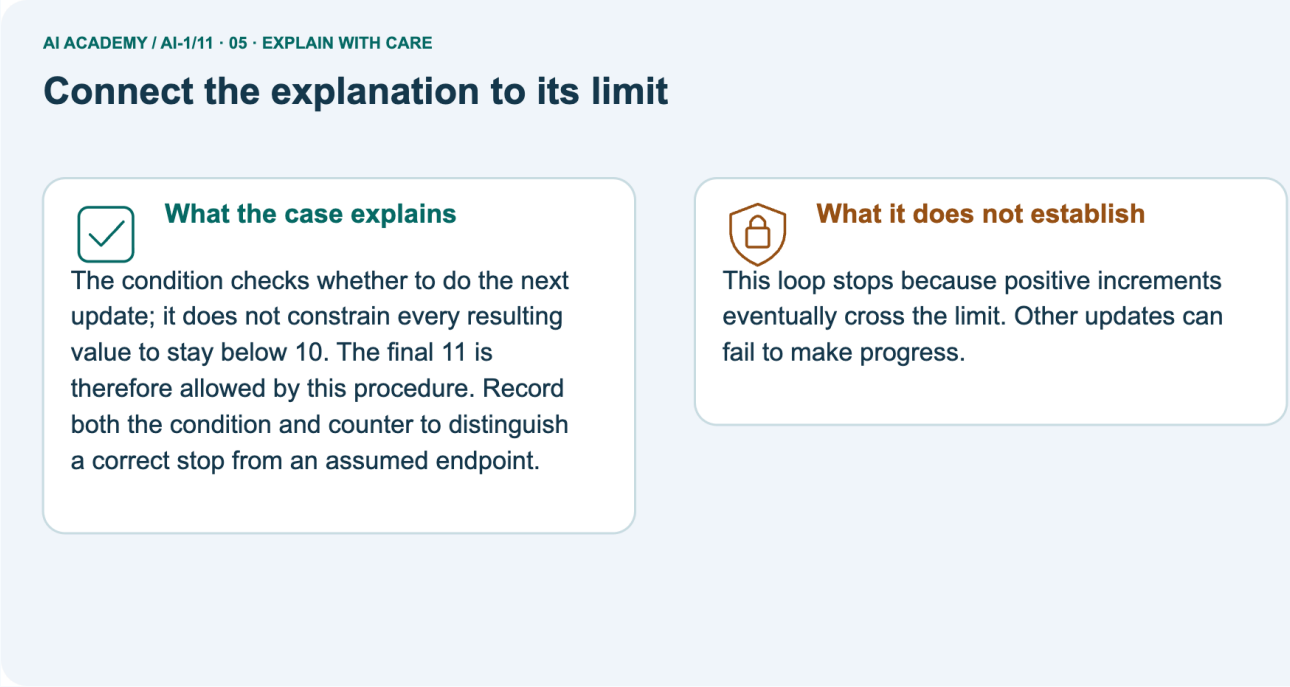
Original case: Final counter 11 after three additions. Changed condition: Start at 10 instead of 2. Expected result: The **loop** runs zero times and reports 10.

Reason: Its first condition is already false, so no addition happens.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10. The final 11 is therefore allowed by this procedure.

Record both the condition and counter to distinguish a correct stop from an assumed endpoint. Limit: This **loop** stops because positive increments eventually cross the limit. Other updates can fail to make **progress**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-1/11 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

Give any repeated sorter step an explicit stopping condition and trace a zero-iteration example.



Trace a fresh start of 9. How many additions occur, and must the report equal 10?

Follow the labelled picture with the matching explanation. Open the vector diagram

Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example. Draw your **input**, deciding step, result and limit. Supported question: Trace a fresh start of 9.

How many additions occur, and must the report equal 10?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.